Preliminary Design Review

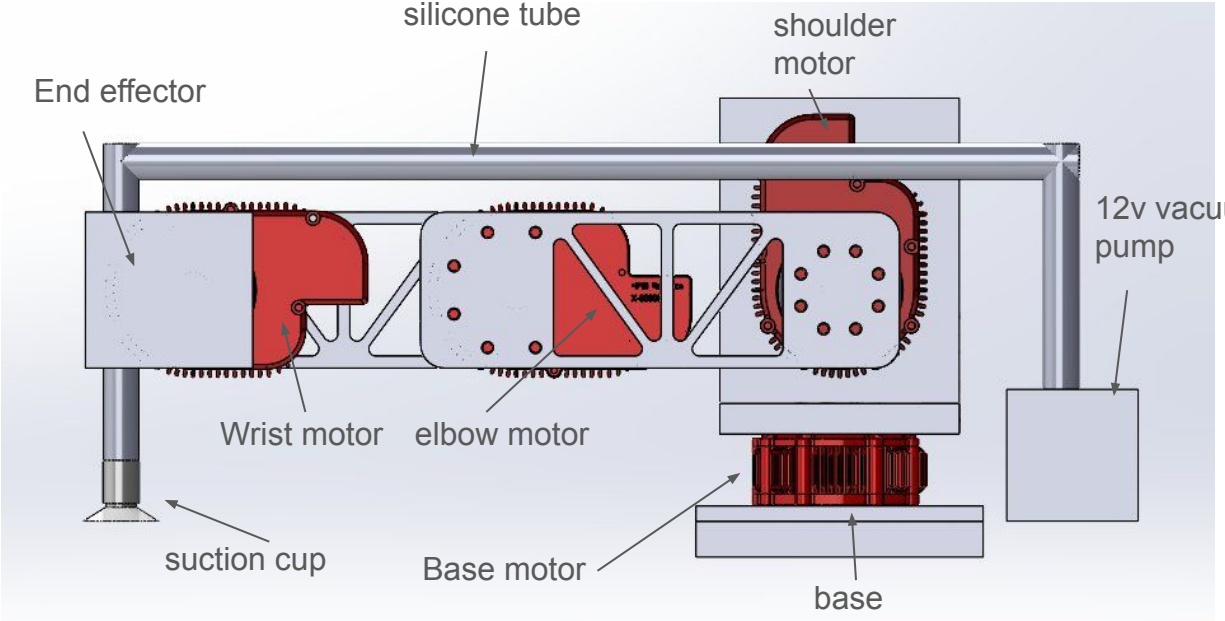
Team GLaDOS

Basic Task: Poker

- Poker dealing and playing robot, which must:
  - Deal out cards to other players
  - Pick up and flip own/community cards
  - Detect own cards without other players seeing
  - Detect community cards
  - Pick up, detect, and count chips
- “Smarts”
  - Human players blocking camera or path of robotic arm → solution: wait for obstacle to be removed and return to home position. Possibly say something using a speaker.
  - Cards stacked on top of each other → solution: spread out cards until each can be distinctly detected and picked up
  - If you drop something → pick it back up!

Overview of Mechanical System

- 4 DOF with approximate 3’ x 3’ workspace (not to scale):



Mechanical System - Purchased Components

- 12v DC vacuum pump:
  - Source: Amazon ($27)
  - May already have the exact same one in the cabinets

- 6mm ID Vinyl Tubing
  - Source: Amazon ($10)

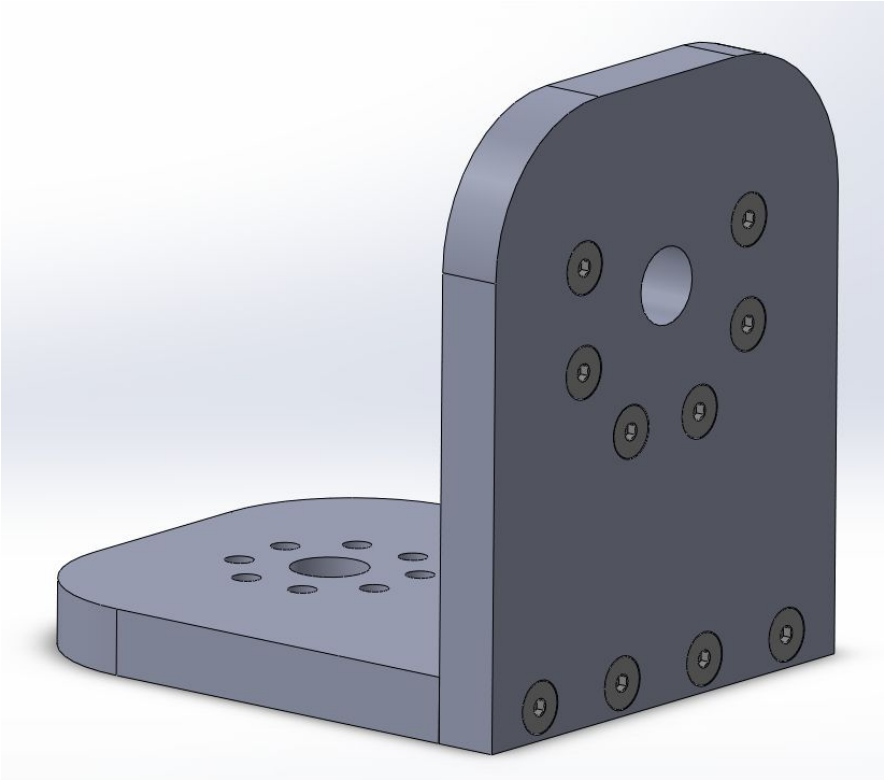
- Suction cup
  - Source: McMaster ($13 each)

- Countersunk 16mm M5 screws
  - Source: McMaster ($9 for 50)

- Aluminum
  - 3”x3’ of ¼ in 6061/3003 aluminum for links
  - 4”x2’ of ½ in 6061/3003 aluminum for base and L-bracket

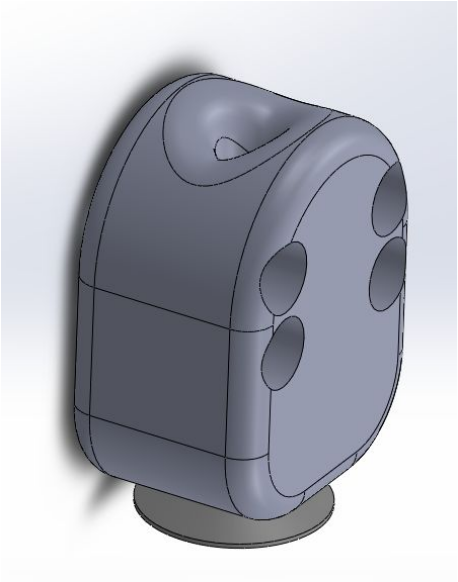
Mechanical System - Machined Components

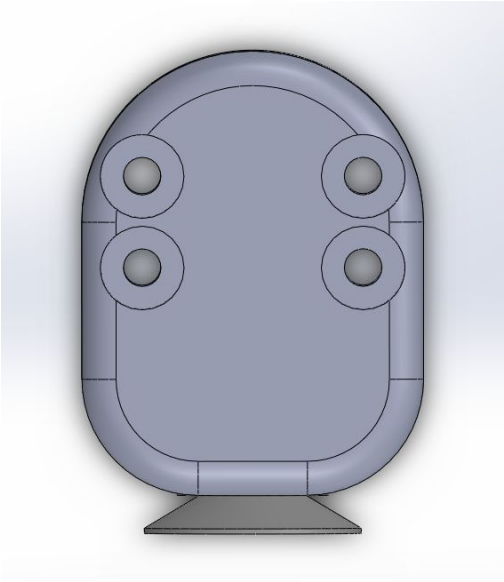
- Base mount:
  - Waterjet 2 parts from ½” aluminum
  - Counterbore and/or tap hardware holes
  - Mill 1 side face to ensure 90°
- L-Bracket
  - Exact same procedure as above
- Links
  - Waterjet 1 part (per link)
  - Drill/tap hardware holes



Mechanical System - 3D Printed Components

- End Effector
  - 3D printed with inserted suction cup
  - Low loads and irregular shape, hence printed





Mechanical System - Approximate Torque Calculations

- Wrist
  - **<1Nm**. Only spins plastic end effector around a 0.04m lever arm.
- Elbow
  - Assuming arm link weighs as much as a 1’x3”x1/4” 6061 bar (400g) centered at midpoint, plus 350g for wrist X5 and 50g for end effector.
  - 0.457m [18”] * (0.400kg + 0.400kg * ½) * 9.81 = **2.7 Nm**
- Shoulder
  - (0.508m [20”] * (0.350kg +0.400kg * ½) + 0.97m [38”] * .600kg) * 9.81 = **8.5 Nm**

Detectors

We will require reliable detection of multiple game pieces, namely:

- The back of cards
- The front of cards including rank and suit
- Poker chips of different colors

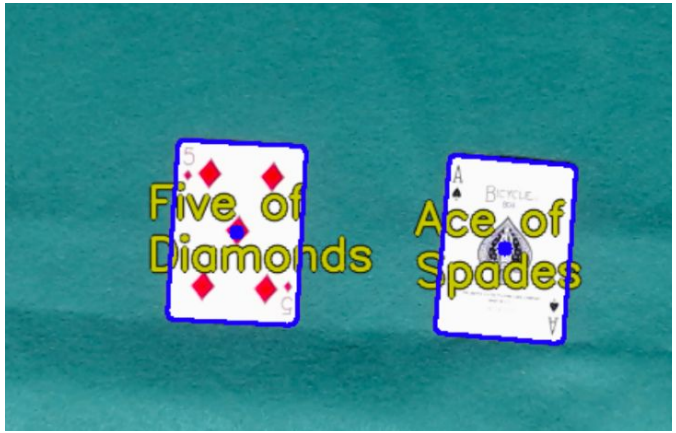
Each of these will have their own unique challenges.

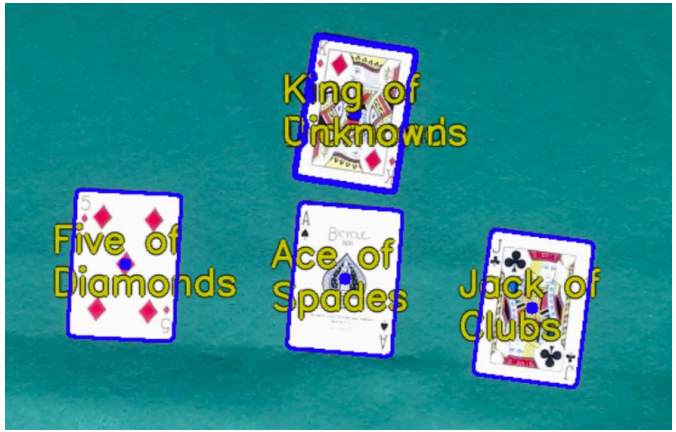
Front of Card Detection

We have a prototype detector working, but there is some noise in the suit detection.

You can see in the second image that the King of Diamonds is being detected some of the time as Diamonds, and some of the time as “unknown”. We’ve noticed that ambient lighting conditions have a large impact on the level of noise.

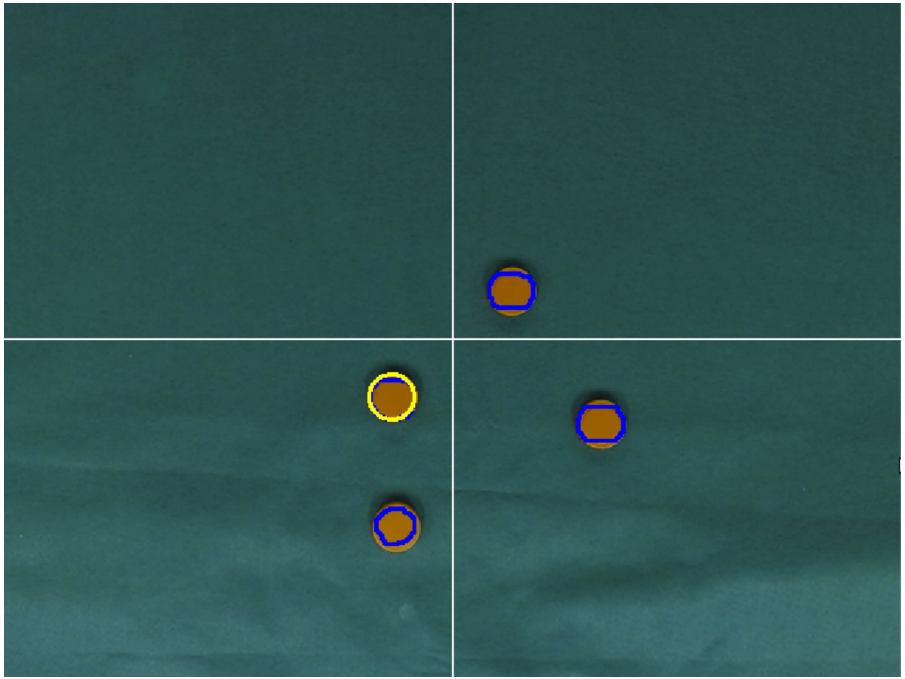
We will also need to be able to tell when several cards are placed down overlapping, which will be a challenge.





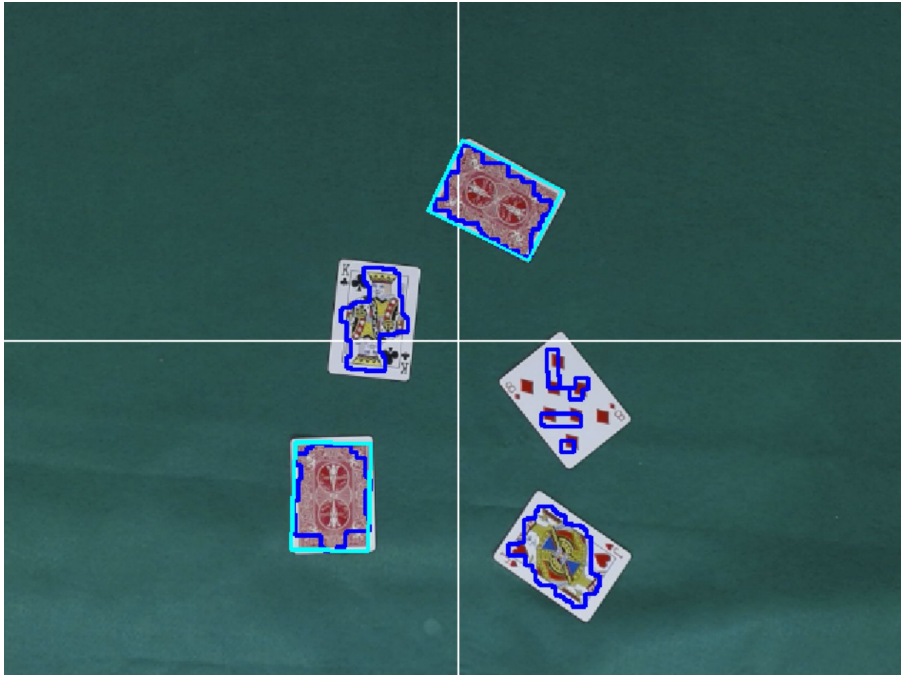
Chip Detection

This is nice and simple! As long as the chips aren't stacked… What should we do if they are stacked? Can we get a depth camera working to count the number of chips in a stack?

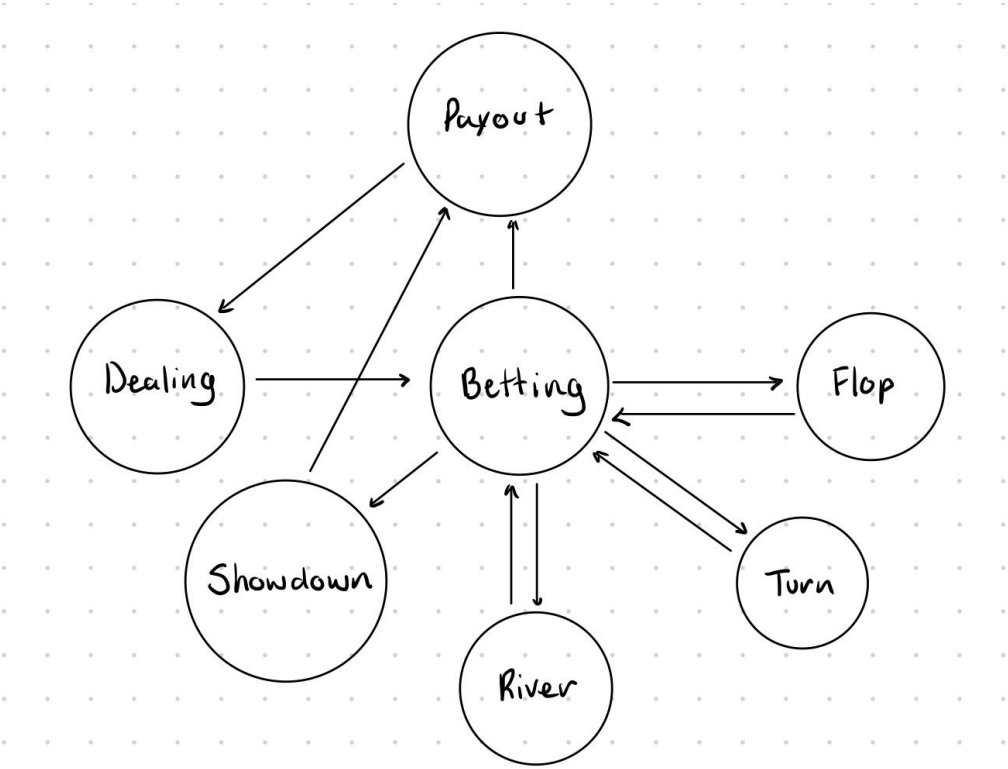


Back of Card Detection

Draw contours around red blobs of a certain area! But again we face the challenge of telling when a number of cards are placed overlapping. Additionally, we may need to be able to tell the height of the deck in order to deal, which we may also have to handle using a depth camera, similar to the height of a stack of chips. (Possibly just go until contact is detected?)



Game States



Interactions

- To declare playing, player puts chips in their player box.
- To bet, player puts chips in pot.
- Dealing is tracked via dealer button (moves after each round).
- Betting is tracked via betting button (to check, move button to next player).
- A player folds by putting cards face down in player box.

Programming Elements

- → states
- → actions
  - ↳ flip card
  - ↳ discard card
  - ↳ "push" chips
  - ↳ "pull" chips

- → detectors
  - ↳ card face
  - ↳ card back
  - ↳ chips
  - ↳ dealer button

- → detected actions
  - ↳ "push" chips
  - ↳ "pull" chips
  - ↳ fold
  - ↳ check

Checks on Cheating

- Verify calls/checks
- Hand detector near stack of cards
- Ensure chip amounts don’t change outside of betting

Reliability Validation

- Drop detection
- Card alignment
- Contact detection

Highest Risk Factors

- The card detector - Current implementation is noisy and sensitive to lighting conditions
- Using suction to flip the cards - This motion will need to be heavily tuned
- Detecting groups of playing pieces - Clusters of cards or stacks of chips will certainly present a hefty detection challenge

Presentation Requirements (Joaquin):

1. Make sure you've thought through manipulating cards. Will you be using some sort of suction? Think about exactly what device you'd like to use (vacuum pump?), ideally with a model number so it can be ordered. Not sure if y'all have separately talked to Günter about this. Same applies for chips, if you're doing that.

2. I'd really recommend you have some sort of tested detector for your cards and have thought through how you'll be telling them apart. Let me know if you need a hand or some thoughts with this.

3. Have clearly defined goals in terms of interactions with the player. Will you be making a hand detector? How will the robot know when it's its turn to move?

4. Think through the failure modes the robot needs. What stuff can it recover from? At what point is it fair to give up?

Presentation Requirements (Gunter 1):

1. What is the basic task?

    1.1. What are the nominal things/motions/actions the system will have to do?

    1.2. What “smarts” should it have? That is, what can go wrong? How will it recover?

    Keep in mind, we’re trying to make the systems reactive. So I’m hoping to see 3-5 distinct reactions to events, and the robot always being aware of whether things are going well or not.

2. What are the implications on the robot:

    2.1. How many DOFS? How are they oriented?

    2.2. What motions can it do, what motions can it not do?

    2.3. Approximate size/workspace.

    2.4. Approximate motor torques are needed at each joint (weights times lever-arms, see below).

    2.5. The CAD or at least a very detailed sketch of the hardware design.

Presentation Requirements (Gunter 2):

3. Gripping/Object manipulation:
    - 3.1. How will you grip/interact with objects?
    - 3.2. Any special features for the fingers or the finger tips?
    - 3.3. Other movements to manipulate objects (pushing/hitting/shoving/etc)?
4. Detection
    - 4.1. What will you have to detect?
    - 4.2. Where will you put cameras, regular or depth?
    - 4.3. How will the detectors work (what do they trigger on)?
    - 4.4. Anything you want to detect non-visually?
5. What are the highest risk items? How do we test/what are backup plans?